

PROJECT DESCRIPTION - STUDENT HANDOUT

BIOSTATISTICS 755, SPRING 2026

1 Overview

In this course, the overall goal is to learn how to do a complete analysis:

1. Decide on the correct statistical model (e.g., linear mixed model, logistic GEE/GLMM, etc.),
2. Develop a covariate strategy that will answer the question of interest (i.e., the hypothesis test for β_3 tests my question of interest),
3. Know how to implement the model in statistical software (e.g., SAS, R),
4. Use data-driven techniques to fine-tune the model and arrive at the final analysis(es). For example,
 - coming up with appropriate adjustment variables,
 - select random effects or covariance structure, and
 - checking model diagnostics (and revising the above as necessary).
5. Clearly interpret the results in general scientific language (i.e., interpreting the value of β_3 so someone with basic scientific knowledge can understand).

Since we learn best by example, you will learn these steps by implementing them on two projects of your design: a midterm and a final project.

1.1 Midterm versus Final Project

The final project requires a final report and a presentation, while the midterm project only requires a presentation. The outcome for the midterm project must be continuous since that's all we will have completed by then. The final project's outcome can be continuous, binary (yes/no), categorical (yes/no/maybe), or count. You can use the same study for both projects, but the study questions should be notably different.

You will work in teams of one or two. You may select with whom you want to work with; working with the same team for both projects is not required.

For the midterm project, we will have not discussed missing data issues, and you will use all available data (even if they have only one observation). As we'll see later in the course, this is an acceptable (if inefficient) way of analyzing data (especially for linear models).

1.2 Topic Selection

The goal of these projects is to apply the methods for longitudinal or clustered data covered in the course. The team will select the project topic. Learning "what questions to ask" is not a goal of this course, but you will have to do some of that here. As far as what data you will use, some possible places to look for data are your advisor, [ADD Health Study](#), [Wisconsin Longitudinal Study](#), or [NHLBI](#). Finding a dataset is part of these projects. This project should not require any data collection.

Some tips on choosing a topic:

- It is best to pick a subject that interests you since it's more likely that it will interest the rest of us.
- Choose a new problem, not something you have already done.
- The data do not have to be related to public health.
- This project aims to implement the methods from the class. Adding topics outside this course's scope usually will not be a good place to put your effort (at least regarding your grade on the project). **You want to spend most of your time ensuring you do and interpret the methods taught in class correctly.** For example, I will have no problem ignoring sampling weights or using a subjective versus objective version of the exposure of interest.
- If your data are longitudinal, the question *should* utilize their longitudinality. Are you looking for a change within the subject over time? Consider asking yourself “could this be answered reasonably well with cross-sectional data?” If so, then your question does not use the longitudinality of the data. For example,
 - **Doesn't use longitudinality:** Is there a difference in BMI between males and females?
 - **Uses longitudinality:** Is the impact of age on BMI different between males and females?
- **Depth over breadth.** Whenever you have questions about what you should be looking at in your project, always choose to investigate one question in-depth over looking at many associations. It's better to say something definitive about one relationship than nothing about three.
- **You don't have to win.** When doing an analysis where there is no relationship between Y and X , the best answer is to find no relationship (even when there “*has to be one*”). Do not repeatedly apply different methods looking for a “better” result. You can ensure your model is correctly specified, but stick to your analysis plan (even when things don't turn out as planned). Again, depth over breadth.

1.3 Types of Projects

The main differentiator in the types of projects is the data used for analysis.

- *Static type*: These projects use data that require straightforward cleaning or transforming of the variables. These are commonly “well-worn” datasets. Note: if your project is a randomized control trial with fewer than four timepoints, this is considered a *static type*.
- *Dynamic type*: In these projects, the data require some meaningful cleaning, transformations, or exploration before use. If you have a dynamic project, you must demonstrate this in your presentation and paper. Provide background on the data and explain the data acquisition process and cleaning (including your solutions).

All projects require exploratory data analyses to determine how variables should be included in models (i.e., should I use X , X^2 , $\log(X)$, or categorize X , same for Y). However, the requirements for the project types slightly differ.

- *Static type*: The team must test multiple model dependence structures (i.e., random effect specifications, covariance types, etc.) and adjustment strategies.
- *Dynamic type*: A single general dependence structure (i.e., baseline random intercepts) can be used, and the investigation into how to adjust it doesn’t have to be as rigorous.

My goal in making this distinction is to respect projects that require significant time for data cleaning. I would love to see a rigorous determination of the best dependence structure and adjustment strategy for all projects. However, if you spend 12 hours cleaning your data, you deserve a break on the analysis side. This is my attempt to do so.

2 Midterm Project

PROJECT ABSTRACT (due on 2/13)

One-page proposals are due in class by the deadline stated above. These proposals should list your team members and briefly state the study’s main goal, your basic approach, and each team member’s role. The purpose of the abstract is to make sure:

- the dataset is appropriate for this class (i.e., longitudinal or clustered data),
- the goals of the project are well defined and
- the outcome and exposure of interest (if any) are feasible.

I don’t grade the abstract. So, do your best to describe your analysis approach. If you don’t know what approach to use, do your best, and we’ll work through it together.

If you don’t think through your analysis before handing in your abstract, the worst thing that will happen is that you won’t get meaningful feedback, which could hurt you down the road. So, think it through, do your best, and make sure you understand the feedback I give you so that the analysis requirements are clear.

ORAL PRESENTATION (3/3 and 3/5)

Each team will give an oral presentation of 15 minutes. The presentation should include all group members. The course website will contain examples of presentations from past years. Your slides should contain all relevant information for the analysis (see the description of the final report for an example outline). The slides and what your group says during the presentation are the majority of your report. **Your slides must include an interpretation of the main coefficient of interest.** This is usually a β coefficient, but could also be a variance term or a least squares means estimate. Statistical code should be emailed separately. Your code is essential, as it specifies what you ran.

GRADING RUBRIC

Team average scores for these projects will be assigned based on the following rubric:

- Quality of the hypothesis generation and the for longitudinal/clustered methods 15%
- Quality of exploratory analysis (figures and tables) 10%
- Model Selection (covariates, covariance, random effects) 20%
- Interpretation of coefficients (even if they're not significant) 10%
- Viability of study conclusions 15%
- Seminar presentation 20%
- Implementation in statistical software 10%

Note: the seminar presentation grade can differ among team members. Implementation in statistical software is judged based on your statistical code (sent separately and not discussed in the presentation).

3 Final Project

PROJECT ABSTRACT (due on 3/27)

See the description from the midterm project.

FINAL REPORT (due on 4/22)

You should thoroughly but concisely report your entire investigation. Include at least:

- Introduction
 - Background/Motivation
 - Current literature on the topic
 - Specific Aims of the project
- Methods
 - description of data variables

- description of how the final analytic sample was obtained
- description of why data are missing and how they were handled
- Statistical methods section
 - Write-up of the analysis plan (exploratory analysis, model section and inferential models)
 - Statement of the statistical model
- Results
 - discussion of exploratory analysis (table/figures)
 - discussion of model section
 - discussion of final results
- Conclusion
 - Relate final results back to the aims of the study
 - Interpretation of coefficients (even if they are not significant).
 - A statement of the implications of your study
 - A discussion of further questions raised by your study
- Meaningful figures and tables that show preliminary patterns in the data.

The main body of the report should include only the end products of any statistical calculations. Statistical jargon will not be well received (i.e., include your findings' interpretation). To quote Richard Feynman, "*Being able to explain or teach a concept shows mastery, but being able to do so simply shows that you truly understand the subject at hand.*"

The paper is to be no more than 5 pages long with standard margins (it may be single-spaced). The 5 pages include all text, but not tables, figures, or bibliography. The order of the report should be the main text, the bibliography, then the tables and figures at the end (statistical code should be emailed separately).

ORAL PRESENTATION (4/21 and 4/23)

Each team will give an oral presentation of 15 minutes. The order of the presentations will be randomized. The course website will contain examples of completed projects and presentations from past years.

Note: the seminar presentation grade can differ among team members.

Table 1: Grading Rubric: Scores for these projects will be assigned based on the following rubric.

Component	Points
Quality of the hypothesis generation	10%
Need for longitudinal/clustered methods	10%
Quality of exploratory analysis (figures and tables)	10%
Model Selection (covariates, covariance, random effects)	20%
Interpretation of coefficients (even if they're not significant)	10%
Viability of study conclusions	15%
Seminar presentation	20%
Readability of the report	5%
Total	100

Table 2: Presentation Tips

Focus Area	Tips
Lead With the Question	Open with a one-sentence research question or hypothesis. State exposure, outcome, and study design upfront so the audience can orient quickly.
Show the Data First	Display key descriptive plots/tables <i>before</i> models (e.g. histograms, spaghetti plots). Highlight missingness, outliers, or clustering to justify the chosen methods.
Explain Modeling Choices	For each model, give a one-line rationale: “We used a mixed model because visits are nested within children.” Specify link functions, covariance structures, and transformations.
Interpret, Don’t Just Report	Translate coefficients into plain language (e.g. “A 0.15 decrease in BMI Z-score per extra daily fruit serving”). Include CIs and put effects in practical terms.
Use Clean Visuals	One key idea per slide; at most two graphics. Label axes clearly ; avoid dense tables—use bold or color to guide attention.
Address Assumptions	Mention diagnostics: residual normality, multicollinearity, convergence. If assumptions fail, note remedies (robust SEs, alternative distribution).
Practice Timing	For a 15-min slot: ≈ 10 content slides + 2 for Q&A. Rehearse aloud; trim until you finish comfortably.
Delivery Matters	Stand where you can see both screen and class; glance at notes, not slides. Speak slightly slower than conversation and use a laser pointer or mouse sparingly. Lastly, smile! We are all rooting for you to succeed.